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Studio Session	11

Part A: URL of the Visualisation

https://minijordy.github.io/FIT3179-A2/

Part B: Domain Description

Describe your domain chosen for DV II

My domain is Australian wine. “Australian Wine: Climate, Regions and Trade” follows wine from the vineyard to the world in four parts: where wine is grown and how warm each region is, what each state grows, what the grapes are worth, and where the finished wine goes, from Australian drinkers to overseas markets.

Describe the Why and Who aspects of your DV II

Why: Australians see wine region names on bottles and heard about China's tariffs on Australian wine (2020–2024), but few know why grapes from some regions are worth more than ten times as much as others, or where most of the wine ends up. This visualisation makes those patterns visible. Who: the average Australian, with no background in statistics or viticulture. Technical terms (e.g. growing season temperature, log scale) are explained where they first appear.

Describe the “What” aspect of your DV II

Public sources, all 14 references are numbered in the page footer:

Wine Australia, National Vintage Report 2026 (regional appendix) – tonnes crushed and estimated value by region and variety: Map 2, heatmap, waffle, grape climate strip, variable-width bar chart, scatterplot. https://www.wineaustralia.com/market-insights/national-vintage-report

Wine Australia, Wine Geographical Indications Australia – wine region (GI) boundaries for all three Australian maps. https://wineaustralia-opendata-wineaustralia.hub.arcgis.com/datasets/2dd4c385f0ed4d109c2e18ae99e819e2

Wine Australia, Australian Wine Export Monitor FY2025–26 – export volume by wine style (waffle chart). https://www.wineaustralia.com/market-insights/australian-wine-exports

Wine Australia, Production, Sales and Inventory Report 2024–25 – wine made, sold and exported (sunburst). https://www.wineaustralia.com/market-insights/australian-wine-production-sales-and-inventory

UN Comtrade, Australia's exports of wine (HS 2204) by country, 2014–2025 – flow map and bump chart. https://comtradeplus.un.org/

Reserve Bank of Australia, Statistical Table F11 – exchange rates to convert Comtrade US dollars to Australian dollars. https://www.rba.gov.au/statistics/historical-data.html

vega-datasets world-110m.json (Natural Earth) – world map for the flow map. https://github.com/vega/vega-datasets

DFAT, Al Jazeera, Export Finance Australia and Austrade – dates of the trade agreements and China's tariffs used in the flow-map notes. https://www.dfat.gov.au · https://www.aljazeera.com · https://exportfinance.gov.au · https://www.austrade.gov.au

Wine Australia, Australia's Wine Future: A Climate Atlas, National Summary (Remenyi et al., 2019) – growing season temperature of each region, 1997–2017 (Map 1, grape climate strip, scatterplot). https://www.wineaustralia.com/climate-atlas

Jones, G. V., Climate change: implications for viticulture and wine production – ideal temperature ranges for Pinot Noir and Cabernet Sauvignon. https://www.infowine.com

University of Adelaide Wine Economics Research Centre, working paper 2021-07 – published temperatures used to check the Climate Atlas values. https://media.adelaide.edu.au/economics/papers/winedoc/winewp2021-07.pdf

Datasets were joined by wine region (GI) name, Comtrade values were converted to Australian dollars with RBA rates, and growing season temperatures were read from the Climate Atlas chart and checked against published values. Total data downloaded by the page is under 1 MB.

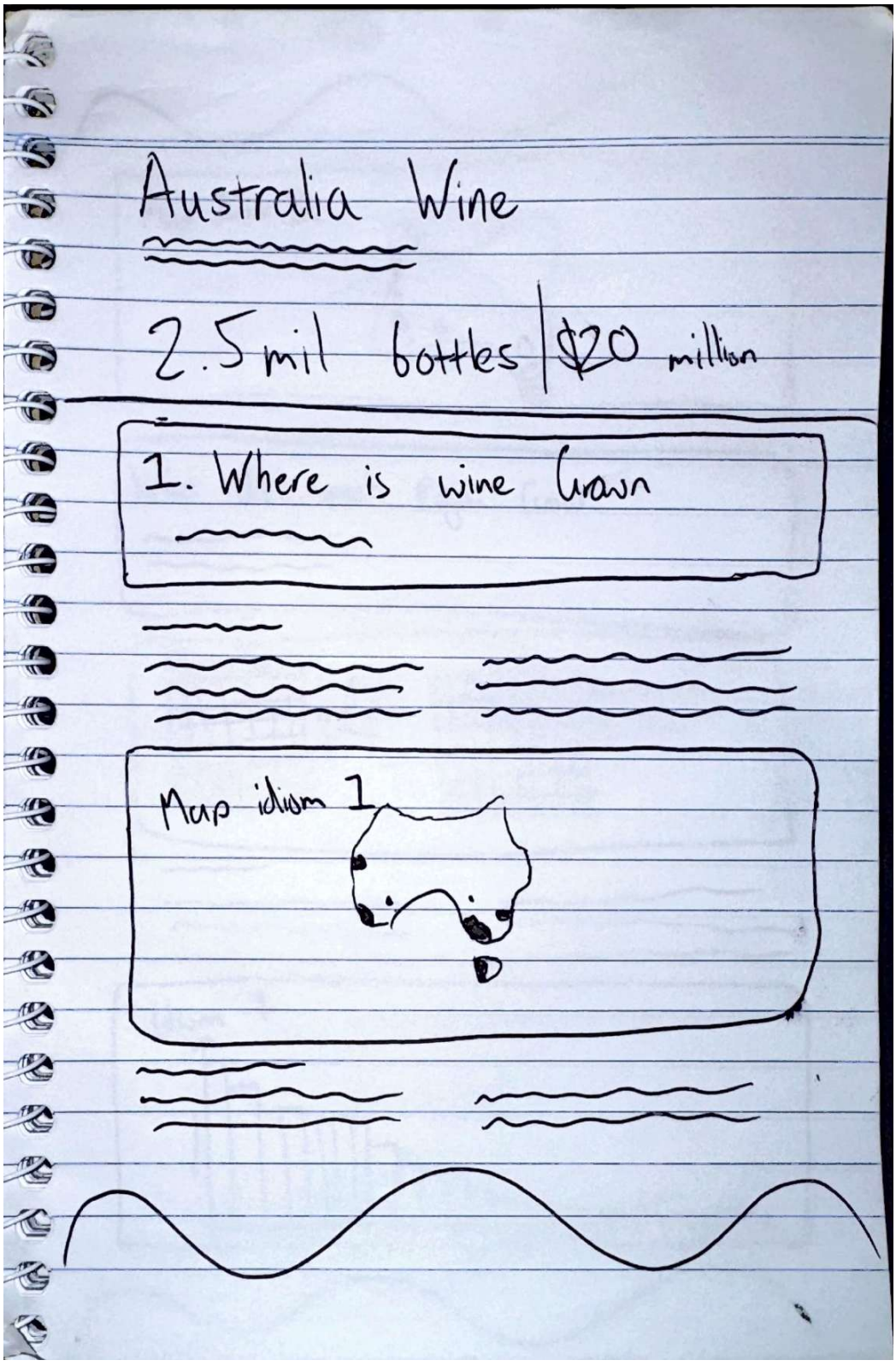
Describe the “How” aspect of your DV II

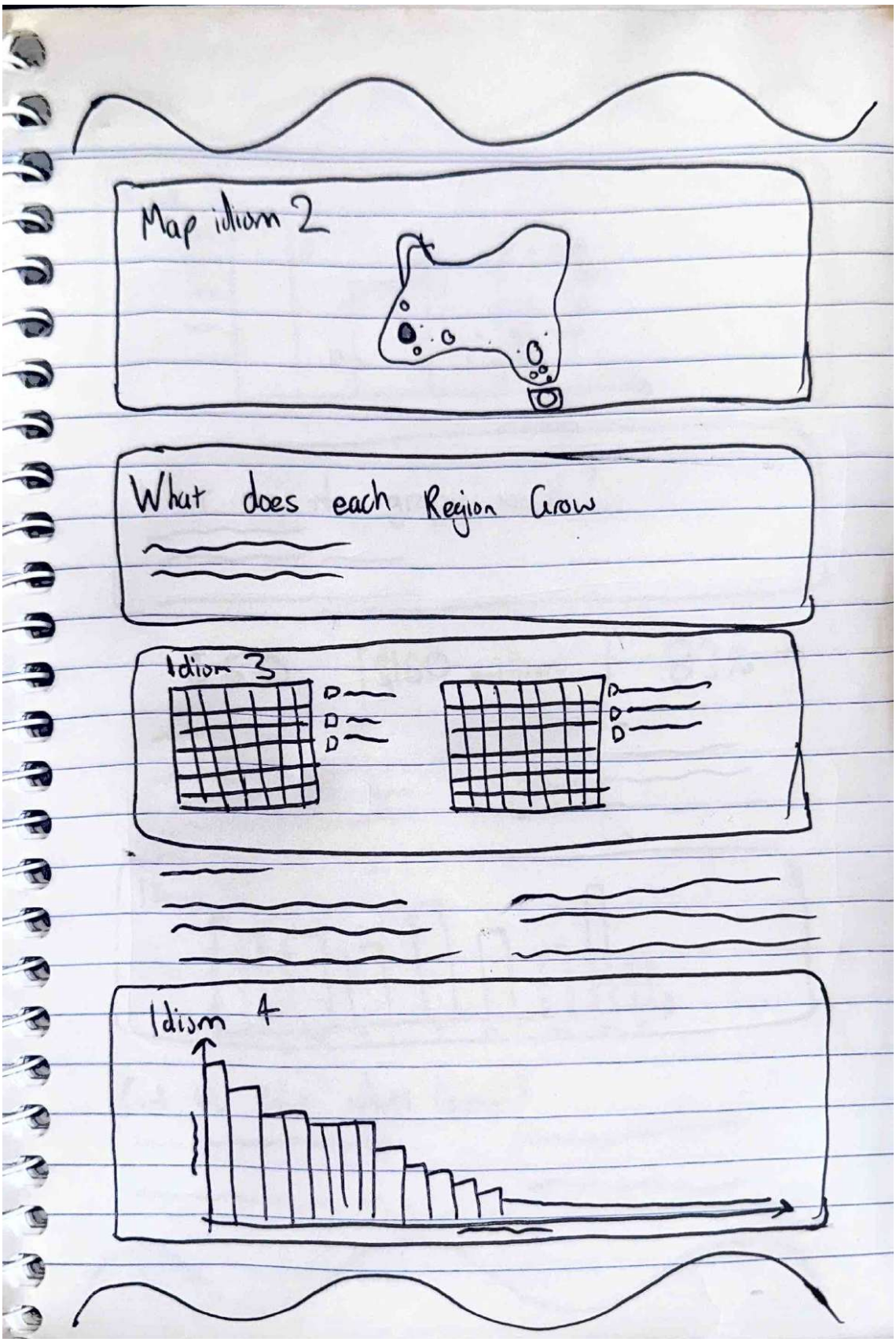
Ten charts were built across eight advanced idioms, including three map idioms (choropleth, proportional symbol map and flow map). Three illustrate the rationale: the variable-width bar chart encodes each region's price per tonne as height and its tonnes crushed as width, so each bar's area is the value of its crush; it shows at a glance that the three hot inland regions grow 72% of the grapes but earn only 36% of the value, which separate price and volume charts would hide. The linked scatterplot and map plot growing season temperature against price on a log scale, and hovering a region in either view highlights it in the other, so readers can connect the climate trend to geography. The flow

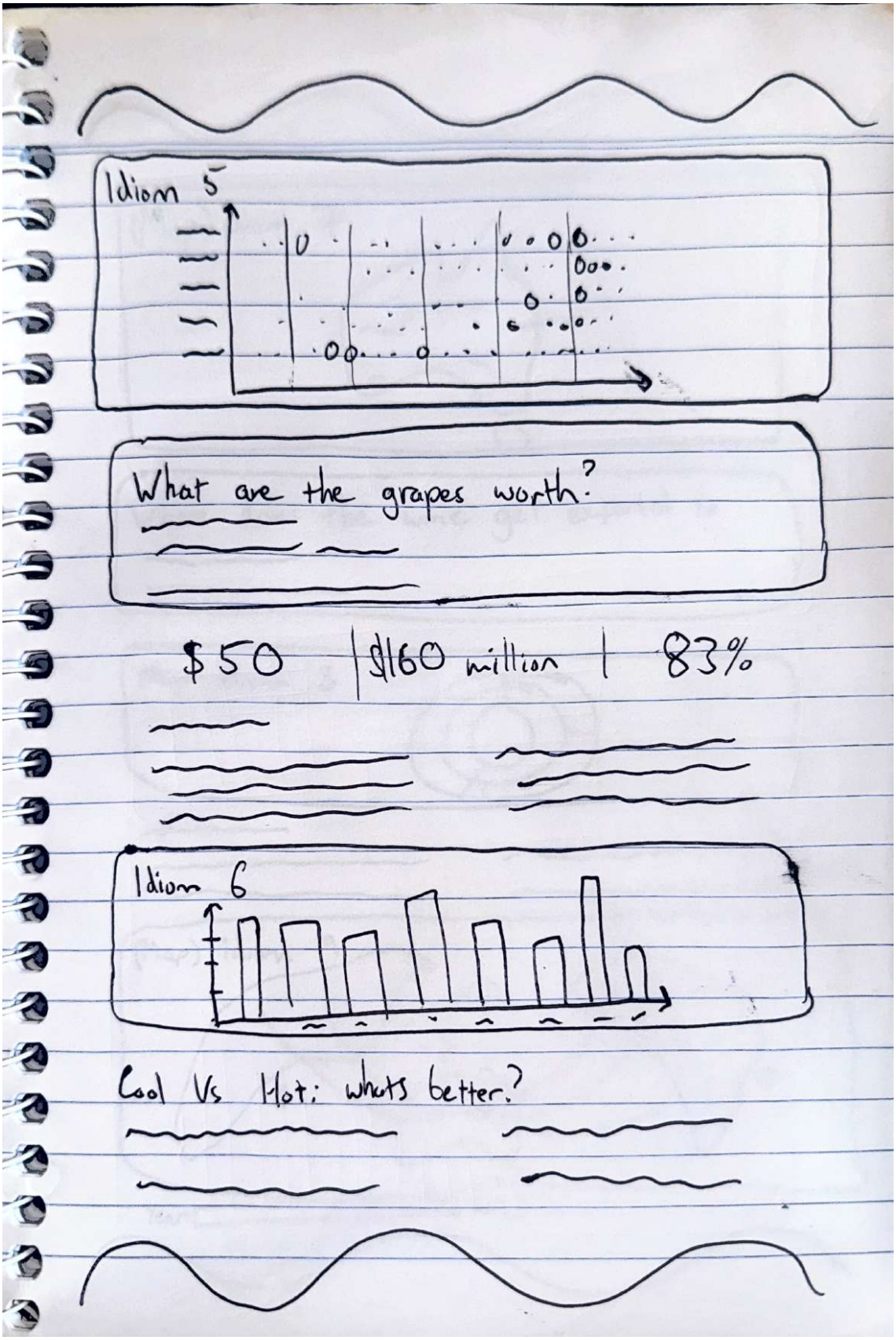
map draws export routes as curved lines whose width encodes value, with a year slider and notes on key trade events, so readers can watch the China market collapse under tariffs and recover. Interactivity is used thoughtfully and purposely, every chart has hover tooltips with exact figures, the grape climate strip highlights a region across every grape, and the bump chart can colour one world region at a time. A consistent palette and theme is used throughout: red, white and rosé for wine styles, a fixed colour for each state, and a merlot theme for structure. Each chart has a short story paragraph beside a how-to-read paragraph, so the reader knows what to look for.

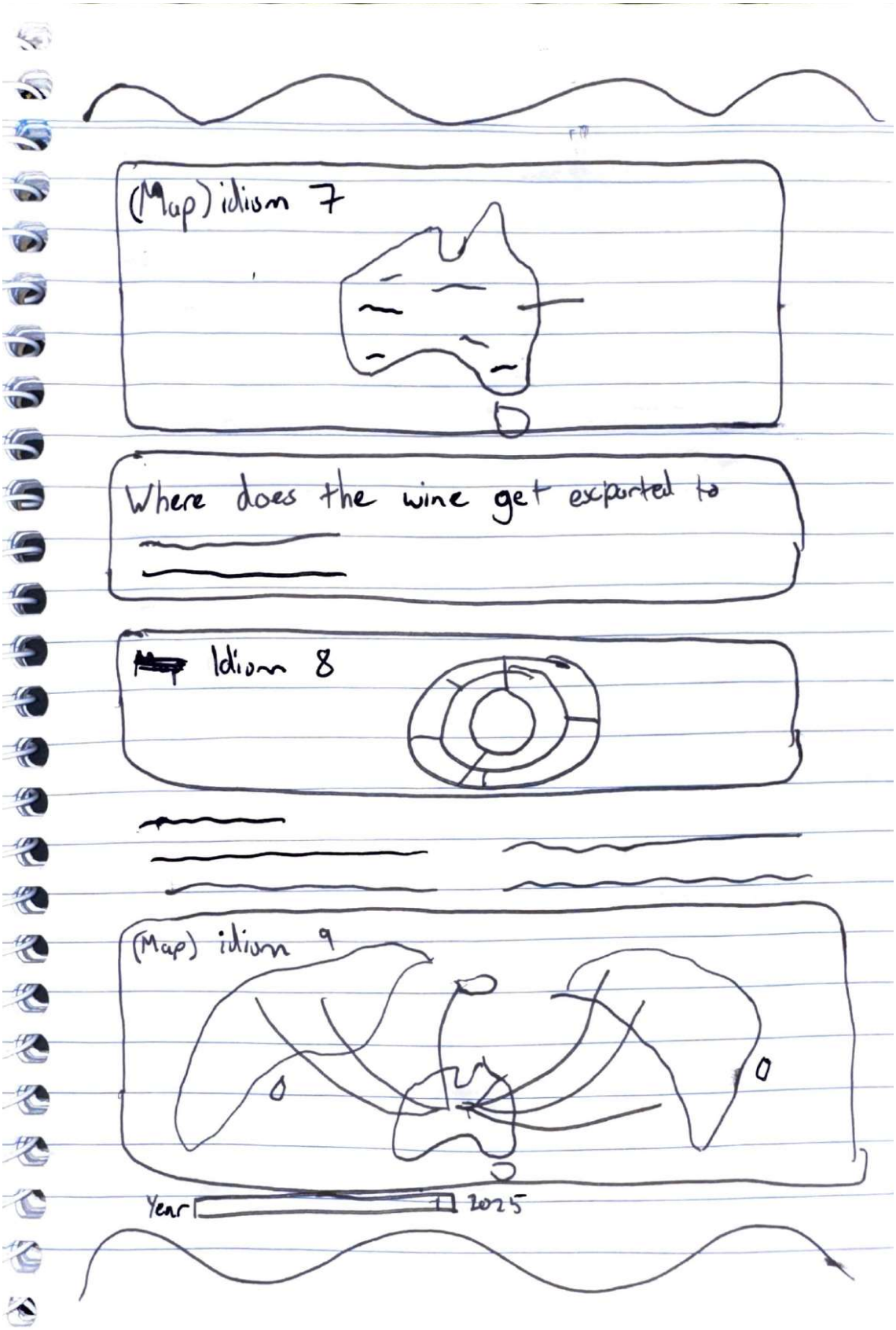
Part C: Sketch

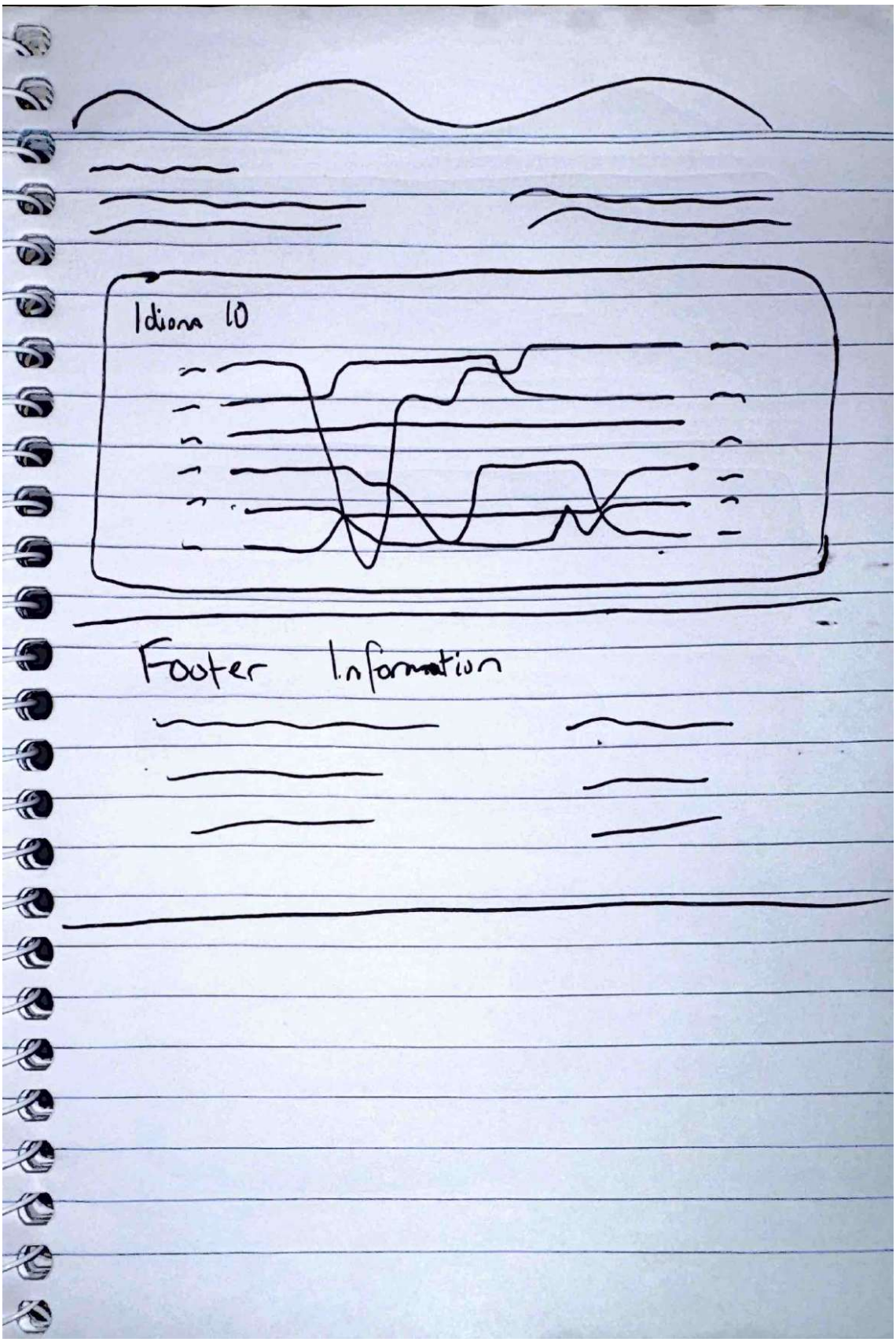
https://minijordy.github.io/FIT3179-A2/sketch.pdf



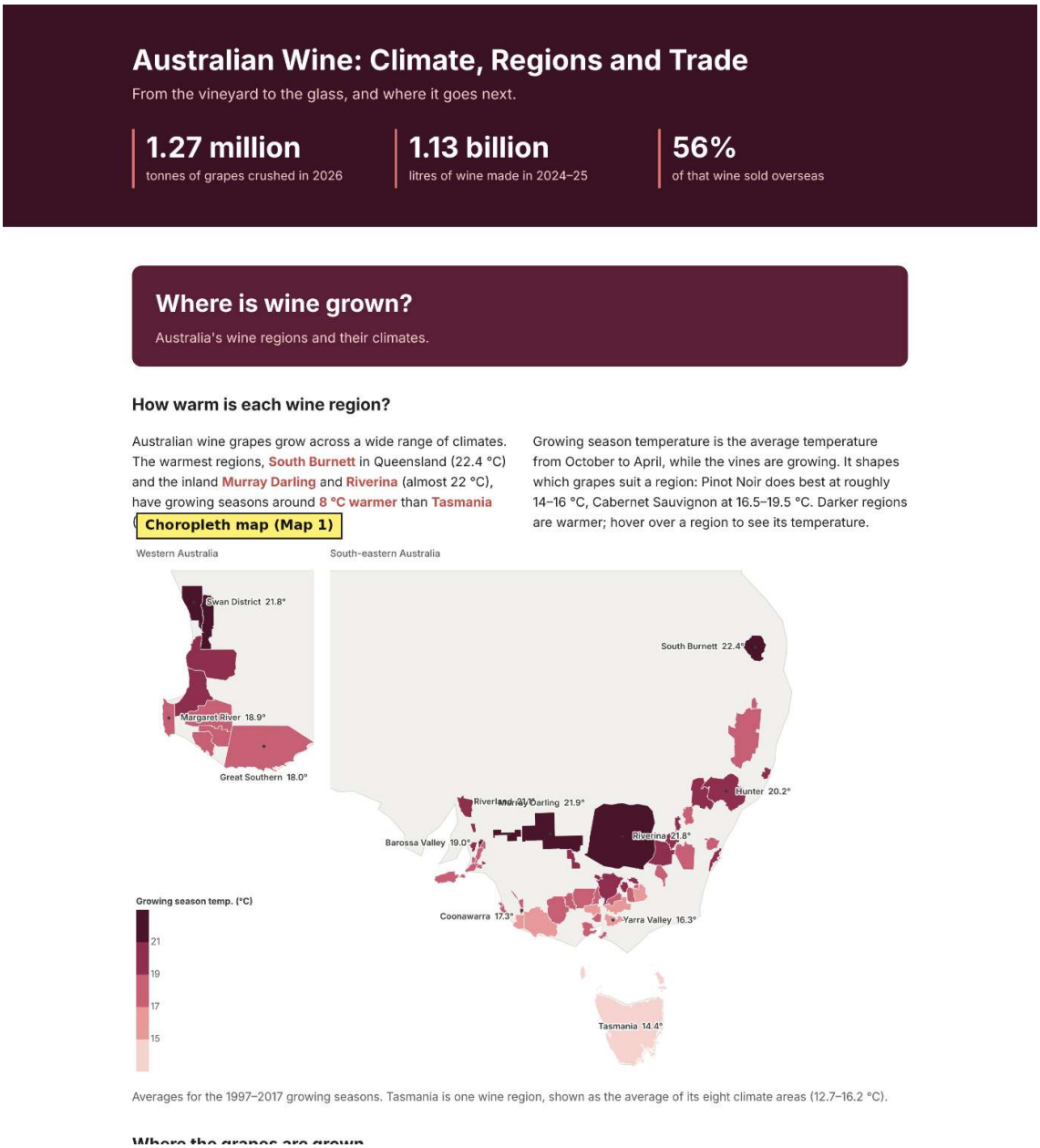

Map idiom 2
What does each Region Grow
Idiom 3
Idism 4


Idiom 5
What are the grapes worth?
$50
$160 million
83%
Idiom 6
Cool Vs Hot: whats better?


(Map) idiom 7
Where does the wine get exported to
Idiom 8
(Map) idiom 9
Year
2025


Idiom 10
Footer Information

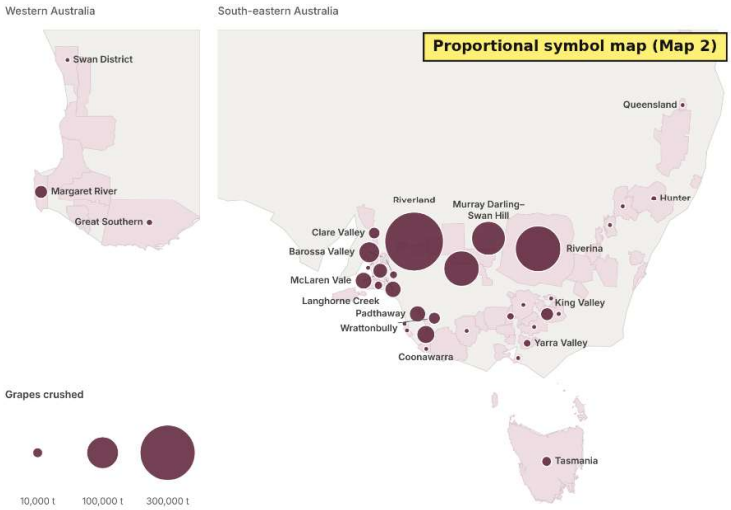
Part D: Full-size screenshot of your DV II



Where the grapes are grown

Most of Australia's grapes come from **three hot inland regions** along the Murray River: Riverland, Riverina and Murray Darling–Swan Hill crushed about **72%** of the grapes reported in 2026, while famous names like the Barossa and Margaret River are much smaller.

Each circle's size shows the tonnes of grapes crushed in that region in 2026.



Very small regions are drawn larger than their true size so they stay visible. Shaded regions without a circle (N/A in the tooltip) crush less than about 1,000 tonnes; Wine Australia doesn't publish figures for them.

What does each place grow?

The grape varieties and wine colours behind those regions.

53%
of the 2026 crush was white grapes

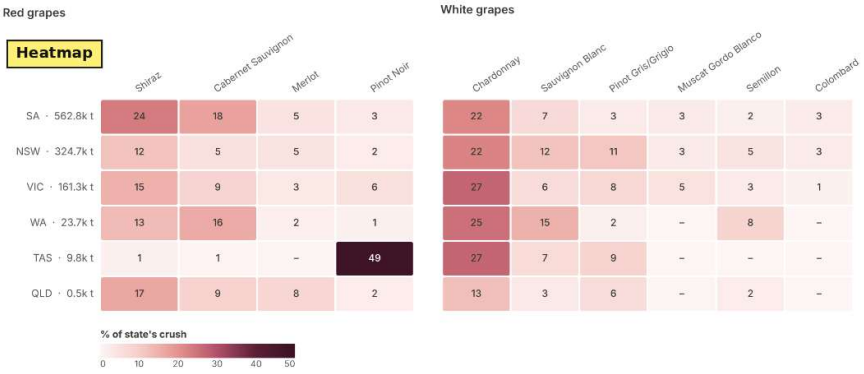
Chardonnay
the most-crushed grape in Australia

35%
fall in the Shiraz crush since 2025

Every state has its own signature

Each state has its own mix of grapes. **South Australia**, the biggest producer, leans on red: Shiraz (**24%**) and Cabernet Sauvignon (**18%**) make up over two-fifths of its crush. **Victoria** and **New South Wales** lean the other way, with Chardonnay their single biggest grape.

Western Australia stands out for Cabernet Sauvignon and Sauvignon Blanc, and **Tasmania**, the coolest state, is **almost half Pinot Noir**, a grape that struggles in hot weather. Each cell shows a grape variety's share of that state's 2026 crush.

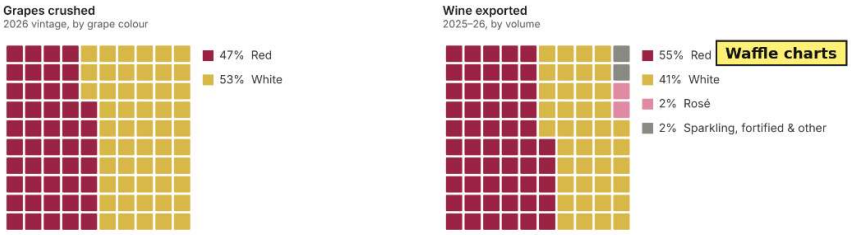


Shows the 10 most-crushed varieties nationally; the rest of each state's crush is other varieties. Queensland is based on only 541 tonnes.

Grown white, shipped red

In 2026, white grapes made up **53%** of the crush, their **highest share since 2012**, because the smaller 2026 crush hit red grapes much harder than white. The Shiraz crush alone fell by **35%** (about 129,000 tonnes), while Chardonnay's barely changed, so Chardonnay overtook Shiraz as Australia's most-crushed grape.

But the wine Australia sends overseas tells the opposite story: **55%** of export volume in 2025–26 was red, and only 41% white. Rosé, sparkling and fortified wines are only a small slice of either. Each square is 1%.

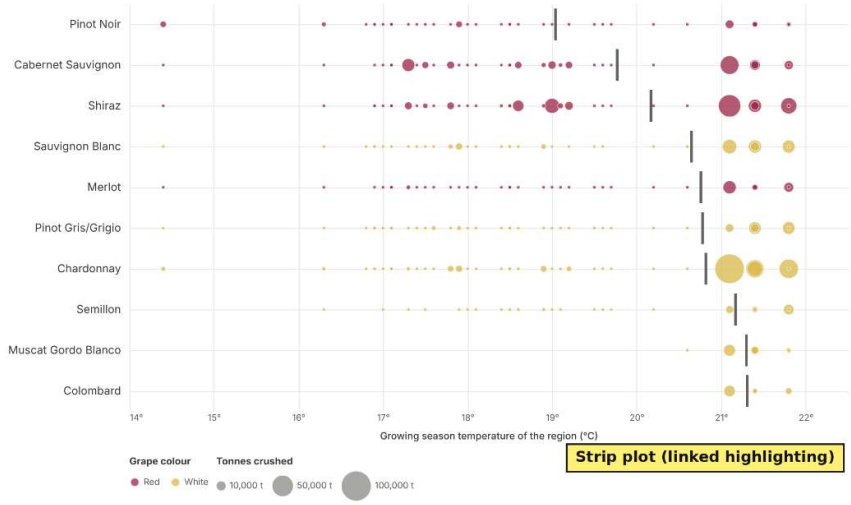


Sparkling, fortified and other wines are grouped because together they make up only 2% of exports.

Every grape has a home climate

Most of Australia's grapes, whatever the variety, come from the hot inland regions, so most of the big circles sit on the right. **Pinot Noir** stands out: it is the main grape of the two coolest regions, **Tasmania** and the **Yarra Valley** (about 7,300 tonnes there, ahead of Chardonnay's 4,400), and has the coolest average (19.0 °C). Bulk grapes such as **Colombard** and **Muscat Gordo Blanco** are almost all inland (21.3 °C).

Each row is one grape. Every circle is a region that grew it in 2026, placed by the region's growing season temperature and sized by the tonnes crushed. The black tick is the grape's average temperature, weighted by tonnes. Hover over a circle to see that region light up in every row.



The same 10 varieties as the heatmap above. Queensland (541 tonnes) is left out because its two regions are reported together.

What are the grapes worth?

Why a tonne of grapes can be worth $300 or $3,000.

$570
average price paid per tonne of grapes

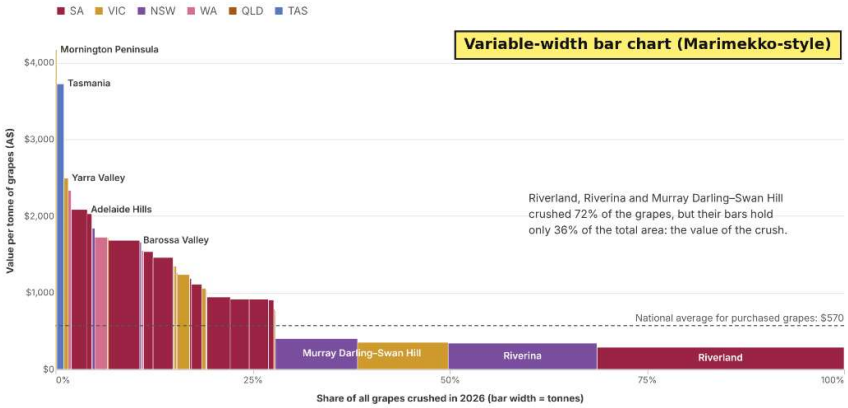
$837 million
total value of the 2026 grape crush

13×
more per tonne in Tasmania than the Riverland

Who gets paid the most for their grapes?

A tonne of grapes from **Tasmania** was worth about **$3,720** in 2026, and from the **Mornington Peninsula** almost **$4,170**. The cheapest grapes come from the regions that grow the most, so the biggest regions earn far less than their size suggests.

Each bar is one wine region. Its height is the value of a tonne of its grapes and its width is how many tonnes it crushed, so the size of each block is the total value of its grapes. The dashed line is the national average price paid for grapes. Hover over a bar for the region's figures.

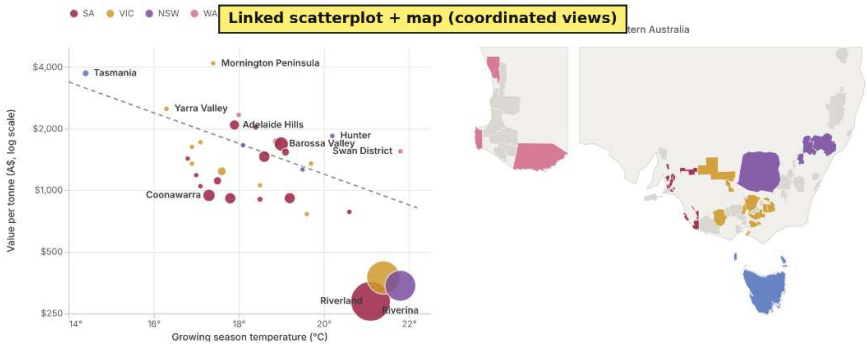


Widths are each region's share of all reported tonnes, so the smallest regions are only slivers. Regions crushing less than about 1,000 tonnes aren't published. Queensland's two regions are reported together.

Do cooler regions earn more?

Mostly, yes. Regions with cooler growing seasons tend to be paid more for their grapes: on the trend line, each extra degree goes with a price about **20% lower**. But temperature is not the whole story. The **Hunter** and **Swan District** are warm yet earn well above the hot inland regions, while **Coonawarra** is cool but sells most of its large crush for under $1,000 a tonne.

Each circle is a wine region, placed by its growing season temperature and the value of a tonne of its grapes, and sized by tonnes crushed. The price axis is a log scale, so each step up doubles the price. Hover over a circle to find the region on the map, or over the map to find it on the chart.



Grey regions on the map have no published crush figures. Queensland is left out because its two regions, with very different temperatures, are reported together.

Where does the wine go?

How much of each year's wine is drunk at home, sent overseas or stored, and which countries buy it.

$2.3 billion
of wine exported in 2025–26

598 million
litres shipped overseas

111
countries bought Australian wine

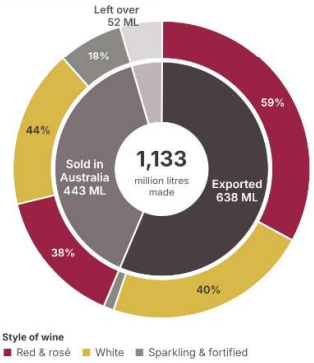
Where did the year's wine go?

Australia made **1,133 million litres** of wine in 2024–25. More than half of it, **638 million litres**, was sent overseas, and most of that was red. Australians drank **443 million litres**, with white wine the favourite at home.

The inner ring shows where the wine went; the outer ring shows what kind of wine it was.

"Left over" is wine made minus wine sold that year; some sales came from older stock.

Sunburst chart



Which countries buy it?

Around **three in every five litres** of Australian wine are sold overseas. **China** is the most valuable market again after its tariffs were lifted, while the **UK** buys the most by volume. Each line's thickness shows the value of wine exported to that market, and the three biggest markets in each year are

Notes in the corner explain the key events in the years they happened. Drag the slider to change the year, or use the menu to show the labels as dollars or as a share of all exports.

Flow map (Map 3)



Year 2025

Show Export value (A$)

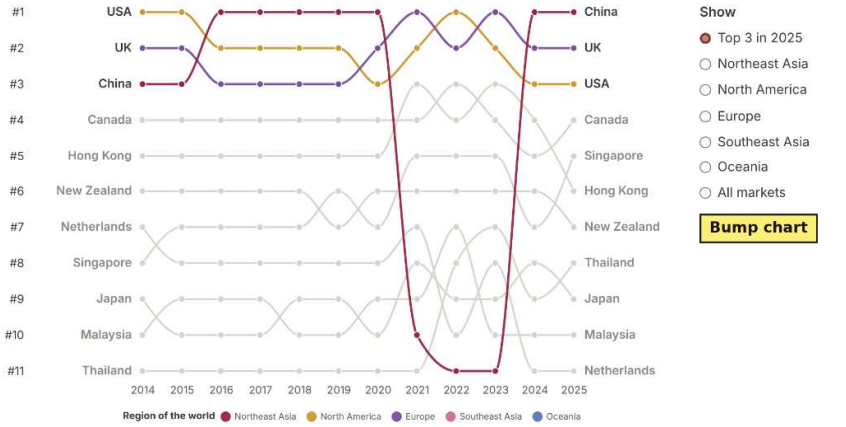
Value and volume tell different stories: in 2025 the UK bought about 200 million litres but paid **under $2 a litre**, while China bought about a third as much and paid **more than $10 a litre**.

Shows the 14 markets that were in Australia's top 10 at some point between 2014 and 2025.

A changing list of customers

China rose to become Australia's most valuable wine market by 2016. When China put heavy duties on Australian wine in late 2020 (up to 218% from March 2021), sales there almost stopped, and China fell out of the top 10 until the tariffs were removed in 2024.

Each line follows one of Australia's 11 main export markets and its rank by the value of wine it bought. The top three markets of 2025 are coloured; use the options beside the chart to colour a region or every market, or hover over a line to follow one market.



Shows every market that was in Australia's top 10 in more than one year, ranked against each other (hover to see each market's rank among all export markets).

Part E: AI Declaration

Model and Description

AI was used throughout this assignment to find and download data, process it into the datasets used by the charts, write and edit the Vega-Lite specifications and helper scripts, critique the layout and charts, and draft narrative text that I then edited. The model used was Anthropic's Claude Opus 5.5.

Usage

AI was used throughout this assignment to assist with many things. It gathered data and created python scripts to wrangle the data and comb through the important parts, as well as writing CSS and HTML code to style the website, as well as creating figures.

Sample Messages

> "Can you critique my visualisation in its current state?"
>
> "to get one more advanced idiom, which chart would you cut/change and what to?"
>
> "I think i just want to cut back some of the text. which paragraphs do you think are least important"
>
> "Now there are some spots with a lot of whitespace that i think i should remove. How do you think i should remove it?"
>
> "I dont really love the new colours, I had some pretty consistent theming in the scheme and now it might feel a bit all over the place."
>
> "I want to add some kind of gradient to try and focus the readers attention towards the middle of the page."